Walmart Hurricane Helene: Demand Surge Analysis
> **DS 2002 Midterm Project** · Malia Solomon, Anika Shah, and Clairine Lou · November 6, 2026

Our notebook is an end-to-end data pipeline and analysis of Walmart sales across Florida stores during September 2024, right before Hurricane Helene's landfall. We clean five data files (six data frames in total, since one of the files splits into two), join them with weather data from a public API, and use the results to answer five business questions about hurricane-driven demand.

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Project Overview

Goal: We are Walmart data science consultants hired to make sense of five data files that don't agree with each other: sales records, store records, a product catalog, distribution center information, and an inventory database. Our goal is to clean the data so it can be trusted, add real weather from an API, answer five questions, and tell Walmart leadership what to do.

Key dates used throughout the analysis:

Period	Dates
Baseline (normal demand)	Sept 1–14, 2024
Pre-landfall surge window	Sept 19–25, 2024
Landfall marker used in charts	Sept 26, 2024

What the notebook does:
1. Loads six raw datasets (CSV files plus a SQLite database).
2. Cleans each one and logs every decision in a decision log and markdown explanations.
3. Consolidates 45 messy SKUs into 20 real products.

4. Pulls daily weather data from the Open-Meteo API.
5. Answers the five analysis questions with graphs and visualizations, and explains each decision for Walmart leadership.

Repository Contents

File	Description
`DS_2002_Project_Working_Doc.ipynb`	Our main notebook with the full pipeline and analysis
`README.md`	This file
`weather_orlando_2024-09-01_2024-09-30.csv`	Cached weather data, created on the first run
`inventory_and_sales.db`	SQLite database, downloaded automatically on the first run

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Data Sources

All of the raw data is loaded directly from our project's GitHub repository:
`https://raw.githubusercontent.com/clairinelou/walmart-hurricane-project/main/data`

Dataset	Source file	Raw shape	Description
`transactions`	`walmart_transactions.csv`	25,562 × 8	Individual sales transactions
`distributions`	`distribution_centers.csv`	5 × 8	Distribution center capacity and costs
`locations`	`store_locations.csv`	14 × 8	Store addresses and coordinates
`product`	`product_catalog.csv`	45 × 7	Product catalog (SKU variants included)
`daily_sales`	`inventory_and_sales.db` → `daily_sales_summary`	3,600 × 5	Daily sales by store and category
`inventory`	`inventory_and_sales.db` → `inventory_levels`	4,320 × 6	Daily inventory by store and SKU

External data:
- **Weather:** [Open-Meteo Historical Weather API](https://open-meteo.com/) (precipitation, max wind speed, mean sea-level pressure; no API key required)
- **Hurricane path:** NHC Forecast Advisory 9 for Helene (issued 1500 UTC Sept 25, 2024), plus the actual landfall point (30.0°N, 83.7°W)

Getting Started

Requirements
- Python 3.9+
- Internet connection (data and weather are fetched at runtime)

Python packages:
```
pandas
numpy
matplotlib
requests
```
`sqlite3`, `pathlib`, `re`, `os`, and `time` are part of the standard library.

Running in Google Colab
1. Upload `DS_2002_Project_Working_Doc.ipynb` to Colab (or open it from Google Drive).
2. Choose **Runtime → Run all**.

Running locally
```bash
# 1. Clone the repo
git clone <your-repo-url>
cd <your-repo-folder>

# 2. Install dependencies
pip install pandas numpy matplotlib requests jupyter

# 3. Launch the notebook
jupyter notebook DS_2002_Project_Working_Doc.ipynb
```

> **Run order matters.** Later cells reuse objects created earlier (for example `sku_map`, `transactions_clean`, `locations_clean`, and the Q1 date variables). Run the notebook from top to bottom.

Pipeline Summary

1. Load
Reads all of the CSVs from GitHub, downloads the SQLite database, and queries both tables.

2. Clean
Each dataset has its own cleaning functions, and every decision is noted in a decision log.

Dataset	Main issues handled
Product catalog	45 SKU variants for 20 products; 27 spellings of 10 categories; missing `unit_cost` (flagged, not filled); unreliable `supplier`, `subcategory`, `package_size` columns (dropped)
Transactions	30 exact duplicates removed; inconsistent store IDs (`fl-412`, `FL_105`); mixed timestamp formats; prices stored as text (`$`); 130 negative quantities set to missing; 884 missing timestamps kept and flagged
Distribution centers	`surge_capacity` stored as text (Miami listed as `~750`), converted to numeric with an approximate flag; Birmingham's missing shipment cost left as missing
Store locations	Mixed ID formats and state labels (`FL`, `Florida`, `Fla.`); ZIP codes stored as text; duplicate Orlando and Kissimmee stores removed (14 → 12 stores); Tallahassee's missing coordinates flagged
Inventory	626 missing `units_on_order` values kept as missing; 355 rows with 4 unmapped SKUs (`UNKNOWN-99`, `LEGACY-PT`, `OLD-BATT-50`, `MISC-001`) kept and flagged
Daily sales	Mixed date formats (`2024-09-12` vs `20240912`); 76 fractional unit totals and 34 negative revenues flagged, not altered

3. SKU consolidation
A shared `sku_map` maps all 45 SKUs to 20 canonical products. Name matching grouped 34 of the SKUs automatically, and we mapped the other 11 by hand. The map is applied to the transactions, product catalog, and inventory.

4. Weather integration
`daily_weather()` calls the Open-Meteo API with retry and backoff for rate limits and server errors, and falls back to a cached CSV if the API is down. Weather is pulled per store using the store's coordinates and local time zone (Pensacola uses Central Time).

5. Decision log
All cleaning steps are combined in a single `full_decision_log` table that records the file, the issue, the decision made, the rows affected, and our reasoning.

Business Questions and Findings

Q1: Demand surge identification
Which categories surged most in the 7 days before landfall versus baseline?

Category	Baseline units/day	Surge units/day	% increase
Breakfast	73.93	376.86	**409.76%**
Emergency	356.57	1,006.43	182.25%
Electronics	69.93	175.86	151.48%
Alcohol	61.57	138.86	125.52%
Health	42.36	95.00	124.28%

Emergency supplies sold the most units, but breakfast grew the most relative to its baseline.

Q2: The Pop-Tarts effect
Four Pop-Tarts SKUs were consolidated into one product (5,005 total units, peaking at 678 units on September 25). The fragmented view makes each SKU look small and could lead to underordering. The consolidated view shows the true demand curve.

Q3: Weather correlation and lag
Spearman correlations compare each store's daily sales (indexed to its own baseline) with local weather, overall and by category. Lag analysis tests whether weather leads or trails purchasing. A positive lag means sales came before the weather; a negative lag means the weather came first.

Indicator	Strongest lag (days)	Correlation at that lag	Correlation at lag 0
Precipitation	−4	−0.53	−0.31
Max wind speed	+4	0.35	0.27
Mean pressure	−6	−0.51	−0.39

Tallahassee (FL-619) is excluded because it has no coordinates.

Interpretation:
- **Same-day relationships are weak to moderate** (none stronger than about ±0.4). Pressure was the strongest indicator, followed by wind, then precipitation. Falling pressure and rising wind go with higher sales, while heavy rain goes with lower sales (people stay home once it arrives).
- **Storm-prep categories responded most.** Emergency and Breakfast (both −0.38 with pressure), Entertainment (−0.29), and Electronics (−0.27) matched the categories that surged in Q1. Snacks showed almost no relationship with any indicator.
- **Wind peaks at a +4 day lag**, meaning sales rose in the days *before* the winds picked up. Pressure is strongest at +1 to +3 days (about −0.42), which points the same way. Shoppers appear to be reacting to the forecast rather than the weather itself.
- **Precipitation peaks at a −4 day lag**, meaning sales were low in the days after heavy rain, which fits shopping dropping off once the storm passed.

- **Treat the pressure peak at −6 days with caution.** It is hard to explain as real customer behavior (the −5 day lag is nearly identical at −0.51).
- **For Walmart:** falling pressure and rising wind forecasts can act as early restocking triggers for Emergency, Breakfast, Entertainment, and Electronics, and sales should be expected to drop after landfall, so avoid over-ordering for those days.

Caveats: these results come from one storm over 30 days, so some correlations may reflect the timeline of the month rather than cause and effect. We explored Orlando further to stay consistent with class material, but it is about 200 miles from the landfall site and only felt the edge of the storm, so correlations may understate the response at landfall. The lags should be confirmed across more storms before being used for store decisions.

Q4: Store-level geographic patterns
Each store's distance to Helene's predicted path was compared with surge size and timing.

- Distance vs. % increase: Spearman r = **0.05** (essentially no relationship)
- Distance vs. days of early warning: Spearman r = **−0.45**
- Outlier stores flagged: **Orlando** and **Miami**

Interpretation:
- Stores closer to the predicted path did **not** surge more. Ten of the 12 stores fell between +137% and +152%. Gainesville, the closest store (137 miles), had one of the smaller surges (+138%), while Miami, the farthest (330 miles), had the largest (+171%).
- Closer stores did **not** surge earlier. Every store except Kissimmee and Daytona Beach began surging on September 16th (10 days before landfall); those two started on the 17th. Every store peaked on September 24th or 25th. The −0.45 correlation comes from that single one-day difference and isn't very meaningful.
- Every store began surging over a week before the September 25th advisory, so shoppers were reacting to statewide awareness of the storm rather than their own store's distance from the path.
- **Orlando** had the weakest surge (+122%), below the roughly +145% predicted from its distance (residual z = −2.26). **Miami** had the strongest (+171%, z = +2.15), possibly because of its low baseline (86.9 units/day) or South Florida's hurricane experience. Tallahassee, excluded from the distance analysis, had the second-largest surge (+160%).
- **For Walmart:** plan hurricane stocking statewide rather than store by store, and start as soon as a storm is perceived as a threat to Florida.

Caveats: only 11 stores have distances and there is a single storm, so one store like Miami can shift the results. Sales reflect

what was sold, not true demand, since stock-outs can hide it. Adding Tallahassee's coordinates is the most useful next step.

Q5: Signal vs. artifact (board games)
The summary database shows a +250% surge in Entertainment (board games and playing cards). We checked whether that is real demand or a data artifact.

Evidence:

Check	What we found
A. Transaction-level breakdown	In the raw receipts, Monopoly rose only **28.3%** (608 → 780 units, baseline vs. surge) and Playing Cards actually fell **1.1%** (829 → 820).
B. Database vs. transactions	For Sept 19–25, the summary database reports **9,761** Entertainment units while the transaction logs show **1,600** (780 Monopoly + 820 Playing Cards), a gap of **8,161** extra units. All **360 of 360** store-days mismatch.
C. Unmapped inventory codes	Four inventory SKUs (`UNKNOWN-99`, `LEGACY-PT`, `OLD-BATT-50`, `MISC-001`) have no valid product mapping. Their on-hand counts swing erratically during the surge window (for example, `LEGACY-PT` jumps to 1,012 units on Sept 20), which is consistent with unreliable legacy records.

Our reading: the +250% looks like a **reporting artifact, not a demand signal**. Our explanation is that legacy SKUs with no valid category mapping were default-grouped into the Entertainment total by the legacy database system, inflating it. The notebook does not directly trace those SKUs into the Entertainment totals, so this is the most likely explanation, not a proven one.

Recommendation: Walmart should **not** stock extra board games for the next storm based on this surge. Ordering against the +250% figure would likely leave Walmart overstocked. That shelf space and budget would be better spent on categories with confirmed demand, such as Emergency supplies and Breakfast items (Pop-Tarts).

Caveats: the transaction file is a sample (roughly 2,000 transactions per store), so it will undercount absolute volume relative to the database. The size of the gap is therefore partly a sampling effect, but the pattern still holds: the raw receipts show little to no board-game surge. Confirming with full point-of-sale data would settle this.

Key Design Decisions

- **Raw data is never modified.** Every cleaning step works on a copy.

- **Flag, don't fill.** Missing and questionable values are flagged (`unit_cost_missing`, `quantity_invalid`, `coordinates_missing`, etc.) rather than guessed.
- **Assertions guard every merge.** Row counts and totals are checked after joins to catch silent duplication.
- **Averages, not totals, for period comparisons.** The baseline covers 14 days and the surge window covers 7, so daily averages keep them comparable.
- **Store-level baselines.** Stores differ in size, so Q3 and Q4 compare each store to its own normal.

Known Limitations

- **Missing timestamps:** 884 transactions have no timestamp and are excluded from date-based analyses.
- **Missing quantities:** 130 transactions have negative quantities that were set to missing and excluded from volume counts.
- **Tallahassee (FL-619):** No coordinates, so it is excluded from weather and distance analyses.
- **Unmapped inventory SKUs:** 355 inventory rows can't be tied to a verified product.
- **Time zones:** Sales timestamps have no time zone attached, so they're assumed to be store-local time.
- **Small sample per store:** Roughly 2,000 transactions per store, so store-level results carry uncertainty.
- **Single month of data:** September 2024 only, with no prior-year comparison.
- **Single storm:** Weather lags (Q3) and geographic patterns (Q4) come from one hurricane and should be validated on others.

Team and Contributions

Name	Role / contributions
Malia Solomon	[e.g., data cleaning, SKU mapping]
Anika Shah	[e.g., weather API, Q3]
Clairine Lou	[e.g., Q1, Q2, visualizations]

License

This project was created as coursework for DS 2002 and is not licensed for reuse or redistribution.

Acknowledgments

- Weather data: [Open-Meteo](https://open-meteo.com/)
- Hurricane track: [National Hurricane Center](https://www.nhc.noaa.gov/)
- Course: DS 2002, [Instructor / University]